

MUZZAMMIL AHMED

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Backend & Full-Stack Engineer | AI Integration | API Design | Cloud Deployment

Software Engineering student (BS, PAF-IAST, 2024–2028) with 4 active internships and a production-grade project portfolio. Experienced in designing and building REST APIs with Python, FastAPI, Node.js, and Express; managing PostgreSQL, MySQL, and MongoDB databases; deploying full-stack applications on Vercel and Render using Docker and CI/CD; and integrating LLM capabilities via OpenAI SDK and OpenRouter. Delivered JWT authentication, Stripe billing, real-time quota enforcement, idempotent API design, and full automated test coverage (Vitest, Playwright, Supertest) across production internship environments.

EDUCATION

Bachelors in Software Engineering

Oct 2024 – May 2028

Pak-Austria Fachhochschule – Institute of Applied Sciences & Technology, Haripur

Coursework: Data Structures & Algorithms, OOP, Database Systems, OS, Software Engineering, Discrete Structures, Linear Algebra, Calculus

WORK EXPERIENCE

FlyRank AI | Backend AI Engineering Intern · Remote

Jul 2026 – Present

- Designed and built production-grade REST APIs using Python, FastAPI, and Flask with Pydantic validation, JWT authentication, SQLite persistence, and Swagger/OpenAPI documentation; followed idempotent API design principles throughout.
- Engineered a resilient web-scraping pipeline (Requests, BeautifulSoup, Pydantic) with caching, retry logic, error quarantine, and idempotent processing to ensure reliable, duplicate-free data ingestion.
- Integrated LLM capabilities into a FastAPI microservice via OpenRouter and OpenAI Python SDK — implemented structured outputs, prompt versioning, cost/token logging, and an LLM kill switch; validated with 8 test cases achieving 100% pass rate.
- Built a Usage Metering & Billing Engine for SaaS: real-time quota enforcement (HTTP 402 on limit breach), accurate monetary calculations with PostgreSQL, and Stripe Checkout + webhook integration for automated subscription billing.

Neurofive Solutions | Full-Stack Development Intern · Remote

Jul 2026 – Aug 2026

- Built component-based frontend applications using React, TypeScript, Vite, and Tailwind CSS; implemented Role-Based Access Control (RBAC) with JWT and bcrypt, protected Node.js/Express API endpoints, and client-side session management.
- Engineered Nestly — a full-stack vacation rental platform (React, Node.js, Express, PostgreSQL/Neon) — featuring server-side booking conflict prevention, dynamic pricing, Cloudinary image pipelines, Recharts host analytics dashboard, and production deployment on Vercel and Render via CI/CD.
- Achieved full-stack test coverage using Vitest, React Testing Library, Supertest, and Playwright — spanning component unit tests, REST API integration tests, and end-to-end user flow automation.

Optimus Automate | Artificial Intelligence Intern · Islamabad

Jun 2026 – Present

- Developing AI-powered automation solutions using Python, FastAPI, and LLM APIs; working on intelligent workflow design, API integration, and production-ready AI application architecture.

Codiuro Technologies | Front-End Developer Intern · Taxila

Jul 2025 – Aug 2025

- Translated Figma designs into responsive, pixel-accurate interfaces using HTML5, CSS3, and JavaScript; enforced cross-browser compatibility and mobile-first standards across Chrome, Firefox, and Edge.

Self-Employed | Academic Tutor — Maths, Physics, CS, Chemistry & English (Part-Time) · Taxila

Aug 2021 – Present

- Delivered structured one-on-one instruction across five subjects to Grades 9–12 students; designed lesson plans that measurably improved exam results and concept retention over four years.

SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, C++, SQL, HTML5, CSS3

Frontend: React.js, Tailwind CSS, Bootstrap 5, Vite, Recharts

Backend: Node.js, Express.js, FastAPI, Python Flask, REST APIs, JWT Auth, Stripe API

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Drizzle ORM, Neon

AI & LLM: OpenAI SDK, OpenRouter, LangChain concepts, Gemini AI API, Prompt Engineering, Gen AI

DevOps & Tools: Git, GitHub, Docker, Vercel, Render, Cloudinary, VS Code, Ubuntu Linux, Swagger/OpenAPI

Testing: Vitest, React Testing Library, Supertest, Playwright

Soft Skills: Problem-Solving, Communication, Mentoring, Team Collaboration, Professional Ethics

Certifications: Gen AI for Everyone · Essential SQL Skills for Data Beginners · Introduction to Artificial Intelligence

PERSONAL PROJECTS

Nestly — Full-Stack Vacation Rental Platform *React · TypeScript · Tailwind CSS · Node.js · Express · PostgreSQL · Cloudinary*

Production-deployed vacation rental app with RBAC and JWT authentication, server-side booking validation preventing date conflicts and race conditions, dynamic pricing engine, Cloudinary image pipelines, and an interactive Host Analytics Dashboard (Recharts) showing real-time revenue and booking statistics. Deployed on Vercel (client) and Render (API) with CI/CD workflows.

[live](#) [repo](#)

Usage-Based SaaS Billing Engine *Node.js · TypeScript · Express · PostgreSQL · Drizzle ORM · Docker · Stripe API* Production-oriented metering and billing backend for SaaS applications — built during internship at FlyRank AI. Idempotent usage APIs using DB constraints to prevent duplicate charges; real-time quota enforcement blocking requests (HTTP 402) on limit breaches; relational data model linking tenants, subscriptions, and usage events; Stripe Checkout and webhook integration for automated subscription billing workflows. [repo](#)

RideFleet Pro — Fleet Management & Ride Booking System *Python Flask · MySQL · HTML5 · CSS3 · JavaScript* Full-stack ride-hailing platform with 3 role-based portals (Admin, Driver, Customer), OTP verification, GPS-based automatic driver assignment, digital wallet with 5 payment methods, fare negotiation, ratings system, and an analytics dashboard. [repo](#)